Introduction

Portfolio optimization is a key concept in finance that focuses on allocating capital across assets to achieve the best possible trade-off between risk and return. This project implements a Monte Carlo simulation-based approach to identify the optimal portfolio using historical stock data.

Objective

The main objectives of this project are:

- To analyze historical stock price data
- To compute return and risk metrics
- To generate multiple portfolio combinations
- To identify the optimal portfolio that maximizes the **Sharpe ratio**

Tools & Technologies

- Programming Language: Python
- Libraries: NumPy, Pandas, Matplotlib
- Data Source: Yahoo Finance API via yfinance

Methodology

Data Collection

- Historical stock data for selected companies (AAPL, MSFT, GOOGL, AMZN, TSLA) was fetched using yfinance.
- Adjusted closing prices were used to account for dividends and stock splits.

Data Preprocessing

- Daily returns were calculated using percentage change of closing prices.
- Missing values were removed to ensure clean data.

Statistical Calculations

- Mean daily returns were computed to estimate expected returns.
- Covariance matrix was calculated to measure the relationship between stock returns.

Monte Carlo Simulation

- 10,000 random portfolios were generated.
- Each portfolio consisted of randomly assigned weights (summing to 1).

For each portfolio, the following were calculated:

- **Expected Annual Return**
- **Volatility (Risk)**
- **Sharpe Ratio**

Optimization

- The portfolio with the maximum Sharpe ratio was selected as the optimal portfolio.
- Corresponding asset weights were extracted.

Visualization

- A scatter plot was created:
  - X-axis: Risk (Volatility)
  - Y-axis: Expected Return
  - Color: Sharpe Ratio
- The optimal portfolio was highlighted on the plot.

Results

- Successfully generated 10,000 portfolio combinations
- Identified the optimal portfolio with:
  - Maximum Sharpe Ratio
  - Balanced risk-return tradeoff
- Determined the optimal allocation for each stock

Key Concepts Used

- Modern Portfolio Theory (MPT)
- Diversification
- Risk vs Return Tradeoff
- Monte Carlo Simulation
- Sharpe Ratio Optimization

Applications

- Investment portfolio construction
- Risk management
- Financial analysis and decision-making
- Algorithmic trading strategies

Limitations

- Assumes historical data predicts future performance
- Risk-free rate assumed as 0% (simplification)
- Does not consider transaction costs or market constraints
- Assumes returns are normally distributed

Conclusion

This project demonstrates how Monte Carlo simulation can be used to explore multiple portfolio combinations and identify the optimal allocation. By maximizing the Sharpe ratio, the model effectively balances risk and return, providing a practical approach to portfolio optimization using real-world financial data.

References

- Financial theory on Modern Portfolio Theory